

Prime Stack Test Case Design

Team: Prime Stack

Application under test: Frost Inventory and Shopping System

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1. Purpose

This document defines the planned test cases for the Frost Inventory and Shopping System. It includes functional, frontend/usability, performance, and security tests, with traceability to the test objectives in the test plan.

2. Functional Test Cases

ID	Feature	Type	Preconditions	Steps	Expected result
TC-FUNC-001	User registration	Normal	Test email does not exist	POST <code>/User/register</code> with valid username, email, password	Status 201, success message, safe user object returned
TC-FUNC-002	User registration	Missing data	None	POST <code>/User/register</code> without email	Status 400, registration rejected
TC-FUNC-003	User registration	Missing data	None	POST <code>/User/register</code> without password	Status 400, registration rejected
TC-FUNC-004	User registration	Missing data	None	POST <code>/User/register</code> without username	Status 400, registration rejected
TC-FUNC-005	User registration	Duplicate	User already exists	POST <code>/User/register</code> with duplicate email	Status 409, duplicate rejected
TC-FUNC-006	User registration	Boundary	None	POST username shorter than 3 characters	Status 400, username rejected
TC-FUNC-007	User registration	Boundary	None	POST password shorter than 6 characters	Status 400, password rejected
TC-FUNC-008	User registration	Invalid format	None	POST invalid email format	Status 400, email rejected
TC-FUNC-009	Login	Normal	Login user exists	POST <code>/User/login</code> with correct credentials	Status 200, login success message
TC-FUNC-010	Login	Error	Login user exists	POST <code>/User/login</code> with wrong password	Status 401; covered by DEF-004
TC-FUNC-011	Login	Error	None	POST <code>/User/login</code> with non-existent username	Status 401; covered by DEF-004
TC-FUNC-012	Login	Missing data	None	POST <code>/User/login</code> with empty body	Status 400; covered by DEF-004
TC-FUNC-013	Authentication status	Normal	No active session	GET <code>/User/status</code>	Status 200, <code>isAuthenticated=false</code>
TC-FUNC-014	Product list	Normal	Products exist or empty table allowed	GET <code>/Product/</code>	Status 200, array returned
TC-FUNC-015	Product details	Normal	Products/suppliers may exist	GET <code>/Product/getALL</code>	Status 200, array returned

ID	Feature	Type	Preconditions	Steps	Expected result
TC-FUNC-016	Product creation	Normal	Supplier id exists	POST <code>/Product/</code> with valid product	Status 200, product created
TC-FUNC-017	Product creation	Duplicate	Product name already exists	POST duplicate product name	Status 409, duplicate rejected
TC-FUNC-018	Product lookup	Normal	Product exists	GET <code>/Product/getByProductId/:id</code>	Status 200, product returned
TC-FUNC-019	Product lookup	Error	Product id does not exist	GET <code>/Product/getByProductId/999999</code>	Status 404, product not found
TC-FUNC-020	Product stock	Normal	Product exists	PUT <code>/Product/decreaseAmount/:id</code> with amount	Status 200, amount updated
TC-FUNC-021	Product deletion	Normal	Product exists	DELETE <code>/Product/:id</code>	Status 200, product deleted
TC-FUNC-022	Supplier list	Normal	None	GET <code>/Supplier/</code>	Status 200, array returned
TC-FUNC-023	Supplier creation	Normal	Supplier name does not exist	POST <code>/Supplier/</code> with name/contact	Status 200, supplier created
TC-FUNC-024	Supplier creation	Duplicate	Supplier name already exists	POST duplicate supplier name	Status 409, duplicate rejected
TC-FUNC-025	Supplier lookup	Normal	Supplier exists	GET <code>/Supplier/:id</code>	Status 200, supplier returned
TC-FUNC-026	Supplier lookup	Edge	Supplier does not exist	GET <code>/Supplier/999999</code>	Status 200, body is null
TC-FUNC-027	Supplier update	Normal	Supplier exists	PUT <code>/Supplier/</code> with updated name/contact	Status 200, supplier updated
TC-FUNC-028	Supplier deletion	Normal	Supplier exists	DELETE <code>/Supplier/:id</code>	Status 200, supplier deleted
TC-FUNC-029	Purchase list	Normal	None	GET <code>/Purchase/getAll</code>	Status 200, array returned
TC-FUNC-030	Purchase creation	Normal	Existing user and product	POST <code>/Purchase/create</code> with products array	Status 201, purchase and purchased products returned
TC-FUNC-031	Purchase creation	Missing data	None	POST <code>/Purchase/create</code> with empty body	Request rejected with validation error
TC-FUNC-032	Purchase by user	Normal	User exists	GET <code>/Purchase/byUserId/:userId</code>	Status 200, array returned
TC-FUNC-033	Purchase sorting	Normal	User exists	GET <code>/Purchase/byUserInAsc/:userId</code>	Status 200, array returned in ascending order
TC-FUNC-034	Purchase sorting	Normal	User exists	GET <code>/Purchase/byUserInDesc/:userId</code>	Status 200, array returned in descending order
TC-FUNC-035	Purchased products	Normal	Purchase exists	GET <code>/Purchase/byPurchase/:purchaseId</code>	Status 200, array returned
TC-FUNC-036	Purchase order parameter	Edge	User exists	GET <code>/Purchase/byUserId/:userId?order=invalid</code>	Status 400, invalid order rejected

3. Frontend and Usability Test Cases

ID	Area	Steps	Expected result
UT-001	Registration form usability	Open Angular app, navigate to login/register, fill registration fields, capture screenshots	Registration form fields are visible/fillable and screenshots are saved
UT-002	Protected client route	Open Angular app and navigate to client dashboard without login	Unauthenticated user is redirected to <code>/login</code>
UT-003	Existing user login usability	Open login page, fill existing credentials, submit login, capture final screen	Login form accepts credentials and request flow completes
UT-004	Product catalog	Login as a seeded client, create a seeded product/supplier, open Products page	Seeded product, price, stock, and supplier name are visible
UT-005	Cart workflow	Login as client, add seeded product to cart, increment/decrement quantity, remove product	Cart reflects item, quantity updates, and empty-cart state
UT-006	Purchase workflow	Login as client, add seeded product to cart, click Buy	Purchase API returns success and success toast appears
UT-007	Profile view	Login as client, open Profile, click Edit	Profile data, purchase log area, and edit modal fields are visible
UT-008	Supplier route protection	Navigate to <code>/Supplier</code> without login	User is redirected to <code>/login</code>
UT-009	Profile edit	Login as client, edit username/email in profile modal, save	Updated username/email are shown on the profile page

Manual usability observations from the teammate review:

ID	Area	Steps	Expected result
UT-MAN-001	Registration visual consistency	Open <code>/login</code> , switch to Register, compare input field backgrounds	All active input fields use consistent styling; recorded as DEF-011
UT-MAN-002	Product discovery/search	Log in as client and inspect dashboard/product discovery workflow	Search or filtering support should be available for larger inventories; recorded as DEF-012

4. Performance Test Cases

ID	Endpoint / workflow	Load condition	Expected result
PT-001	GET <code>/User/status</code>	5 to 50 virtual users	Status 200, response under 200 ms for individual check
PT-002	GET <code>/Product/</code>	5 to 50 virtual users	Status 200, response under 300 ms for individual check
PT-003	GET <code>/Product/getALL</code>	5 to 50 virtual users	Status 200, response under 400 ms for individual check
PT-004	GET <code>/Supplier/</code>	5 to 50 virtual users	Status 200, response under 300 ms for individual check
PT-005	POST <code>/User/login</code>	5 to 50 virtual users	Status 200, response under 500 ms for individual check
PT-006	Overall API	Ramp 5 VUs for 30s, 20 VUs for 1m, 50 VUs for 30s, ramp down 30s	95th percentile response time below 500 ms and error rate below 10%

5. Security Test Cases

ID	Risk area	Attack / input	Expected result
ST-001	SQL injection login	Username/password: ' OR '1'='1	Login rejected, no user data returned
ST-002	SQL injection URL parameter	/Product/getByProductId/1' OR '1'='1	Request rejected or sanitized; database contents not exposed
ST-003	XSS registration	Username: <script>alert("xss")</script>	Input should be rejected; covered by DEF-007
ST-004	Broken access control	GET /User/profile without session	Status 401
ST-005	Mass assignment	Register body includes role: "admin"	User should not be granted admin role; covered by DEF-005
ST-006	Oversized payload	100,000-character username/password	Server does not crash
ST-007	Unauthorized data access	GET /User/users without session	Should return 401; covered by DEF-006
ST-008	Security headers	ZAP scan of http://localhost:3000/Product/	X-Content-Type-Options: nosniff should be present; covered by DEF-008
ST-009	Information disclosure	ZAP scan of missing route such as /robots.txt	X-Powered-By should be suppressed; covered by DEF-008
ST-010	CSP configuration	ZAP scan of missing route such as /robots.txt	CSP should define frame-ancestors and form-action ; covered by DEF-008
ST-011	Authenticated session access	Register/login a client in k6, then GET /User/profile with session cookie	Status 200 and returned profile matches logged-in user

6. Traceability Matrix

Requirement / objective	Test cases
User authentication works and validates input	TC-FUNC-001 to TC-FUNC-013, UT-001 to UT-003, UT-007, UT-009, ST-001, ST-004, ST-011
Product inventory can be managed	TC-FUNC-014 to TC-FUNC-021, UT-004
Suppliers can be managed	TC-FUNC-022 to TC-FUNC-028, UT-004, UT-008
Purchases and cart actions work	TC-FUNC-029 to TC-FUNC-036, UT-005, UT-006, UT-007
Application responds under load	PT-001 to PT-006
Security risks are identified	ST-001 to ST-011